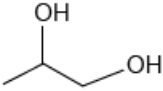
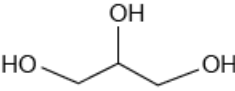


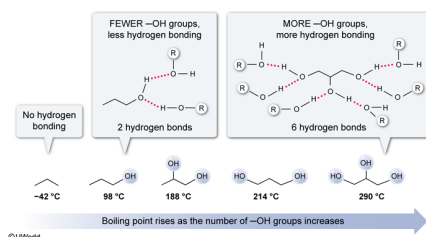
Which of the following compounds has a boiling point that is more than 30 °C higher than that of 1,3-propanediol, HOCH₂CH₂CH₂OH?

- ☐ A.  [1%]
- ☐ B.  [2%]
- ☐ C.  [3%]
- ✓ ☒ D.  [92%]

Correct

93% answered correctly

Explanation:



Intermolecular forces cause attractions between molecules. Structures with a **greater** extent of **intermolecular forces** have **higher boiling points** because more energy is required to overcome these attractions between molecules during the **phase change** from a liquid state to a gas state. Hydrogen atoms covalently bonded with nitrogen, oxygen, or fluorine atoms form **polar bonds** with partial charges producing **dipoles**. Attractions between partial charges of opposite signs generate a significant intermolecular force known as **hydrogen bonding**. This term does not refer to the covalent chemical bond between hydrogen and another atom but refers instead to the **attractions** between the hydrogen bond dipoles.

When comparing similar molecular structures, molecules with a greater number of hydroxyl groups (–OH groups) have more sites capable of hydrogen bonding. As the number of –OH groups increases, more energy is required to overcome the additional attractions from hydrogen bonding and the boiling point rises. Accordingly, only 1,2,3-propanetriol (3 –OH groups) has a higher boiling point than 1,3-propanediol (2 –OH groups).

(Choice A) Because 1-propanol has one fewer –OH group and forms fewer hydrogen bonds than 1,3-propanediol, 1-propanol has a *lower* boiling point than 1,3-propanediol.

(Choice B) 1,2-propanediol has the same number of –OH groups as 1,3-propanediol (different only by position), but the boiling point of 1,2-propanediol is 30 °C *lower* because **branched structures** experience fewer **London force** interactions than comparable straight-chain molecules.

(Choice C) Propane has a boiling point markedly lower than 1,3-propanediol because propane lacks –OH groups and is incapable of forming hydrogen bonds.

Educational objective:

Structures with a greater extent of intermolecular forces have higher boiling points because more energy is required to overcome the attractions between molecules. Among similar structures, the boiling point tends to increase as the number of hydroxyl groups increases due to a greater extent of hydrogen bonding interactions.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401780
System:
Thermodynamics, Kinetics & Gas Laws